INCH-POUND

MIL-W-53122(ME)
31 January 1992

MILITARY SPECIFICATION

WATER PURIFICATION UNIT

REVERSE OSMOSIS, 3,000 GPH

This specification is approved for use within the USA Belvoir Research, Development, and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification defines a trailer mounted reverse osmosis water purification unit hereinafter referred to as "ROWPU", having a water production rate (see 6.7.d) of 60,000 gallons per day (GPD) with minimum average flow rate of 3,000 gallons per hour (GPH) when processing fresh/brackish water and 40,000 GPD with a minimum of 2,000 GPH when processing salt water (all water flow corrected to 77 °F by using appropriate RO element manufacturer normalization chart).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4610

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SPECIFICATIONS

FEDERAL

ZZ-H-451	- Hose, Fire, Woven-Jacketed Rubber or Latex or Rubber-Coated Fabric-Lined, with Couplings.
ZZ-H-561	- Hose, Rubber, and Hose Assemblies, Rubber, Smooth Bore, Water Suction and Discharge.
PPP-B-601	- Boxes, Wood, Cleated Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-1055	- Barrier Material, Waterproofed, Flexible.
PPP-P-40	- Packaging and Packing of Hand Tools.
PPP-T-76	- Tape, Pressure-Sensitive Adhesive Paper, (for Carton Sealing).

MILITARY

MIL-P-116	- Preservation, Methods of.
MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-V-173	- Varnish, Moisture and Fungus Resistant (for Treatment of Communication, Electronic, and Associated Equipment).
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting Materiel.
MIL-L-2104	- Lubricating Oil, Internal Combustion Engine, Tactical Service.
MIL-F-3541	- Fittings, Lubrication.
MIL-G-3859	- Grease Guns, Hand-Operated, Lever, Push and Screw Type.
MIL-W-6858	- Welding, Resistance, Spot and Seam.
MIL-B-7883	- Brazing of Steels, Copper, Copper Alloys, Nickel Alloys, Aluminum and Aluminum Alloys.
MIL-P-10603	- Pumps and Pumping Units, Centrifugal, Power-Driven, for Water, Packaging of.
MIL-P-19834	- Plates, Identification, Metal Foil, Adhesive Backed.
MIL-L-21260	- Lubricating Oil Internal Combustion Engine Preservative Break.
MIL-C-22750	- Coating, Epoxy Polyamide.
MIL-G-23827	- Grease, Aircraft and Instrument, Gear and Actuator Screw, NATO CODE G-354, Metric.
MIL-G-28554	- Generator Sets, Mobile Electric Power, Packaging of.
MIL-L-46167	- Lubricating Oil Internal Combustion Engine Arctic.
MIL-C-46168	- Coating, Aliphatic Polyurethane, Chemical Agent Resistant.
MIL-P-53022	- Primer Epoxy Coating Corrosion Inhibitive Lead and Chromate Free.
MIL-P-53030	- Primer Coating Epoxy Water Reducible Lead and Chromate Free.

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MIL-C-53039	- Coating Aliphatic Polyurethane Single Component.
MIL-C-53072	- Chemical Agent Resistant Coating (CARC) System Application Procedures and Quality Control Inspection.
MIL-P-53084	- Primer Cathodic Electrodeposition CARC.
MIL-V-62038	- Vehicles, Wheeled, Preparation for Shipment and Storage.

STANDARDS

FEDERAL

FED-STD-313	- Material Safety Data Sheets.
FED-STD-595	- Colors used in Government Procurement.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of U.S. Military Property.
MIL-STD-209	- Sling and Tie-down Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-454	- Standard General Requirements for Electronic Equipment.
MIL-STD-461	- Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-462	- Electromagnetic Interference Characteristics Measurement Of.
MIL-STD-810	- Environmental Test Methods and Engineering Guidelines.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking and Waterproofing; with Appropriate Test Methods.
MIL-STD-1472	- Human Engineering Design Criteria for Military Systems, Equipment and Facilities.
MIL-STD-1474	- Noise Limits for Military Materiel.
MIL-STD-1791	- Design for Internal Aerial Delivery in Fixed Wing Aircraft.
MIL-STD-6866	- Inspection, Liquid Penetrant.
MIL-STD-11991	- Electrical, Electronic and Electro-Mechanical Equipment, Guided Missile and Associated Weapon Systems, General Standard for the Design of.
MS35844	- Plug, Machine Thread, Magnetic (Drain).
MS49006	- Plug, Pipe, Magnetic-Headless, Aluminum, Iron or Steel.

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(Unless otherwise indicated, copies of federal and military specifications and standards are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWING

TA13229E0000	- Water Purification Unit, Reverse Osmosis, Trailer-Mounted, 3,000 GPH.
13229E1378	- Overpack Assembly, 3000 GPH ROWPU.

(Copies of drawings required by contractors in connection with specific acquisition functions should be obtained from the USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-FSQ, Fort Belvoir, VA 22060-5606.)

INTER-GOVERNMENTAL MARTIME ORGANIZATION

International Convention for Safe Containers (SCS)

(Applicatioon for copies should be addressed to the Commandant, (G-MVI), US Coast Guard, 2100 Second Street, SW Washington, DC 20593.)

CODE OF FEDERAL REGULATIONS

Title 21, Food and Drug Additives, Parts 170 to 199.
Title 29, Chapter XVII, Part 1910.
Title 49, Parts 170 to 190.

(Application for copies should be made to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

DEPARTMENT OF ARMY TECHNICAL BULLETIN

TB MED 577	- Sanitary Control and Surveillance of Field Installation Water Supplies.

(Application for copies should be made to the Commander, US Army AG Publications Center, 2800 Eastern Boulevard, Baltimore, MD 21220.)

PUBLICATIONS

(C) ABCA Agreement (QSTAG 244)	- Nuclear Survivability Criteria for Military Equipment. (U)
(C) ABCA Agreement (QSTAG 747)	- Nuclear Comtamination Survivability Criteria for Military Equipment. (U)

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(Application for copies of the QSTAG should be addressed to the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/IEEE 315 - Graphic Symbols for Electrical and Electronic Diagrams (Including Reference Designation Class Designation Letters).

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR NON-DESTRUCTIVE TESTING (ASNT)

ASNT-TC-1A Recommended Practice - Non-Destructive Testing Personnel Qualifications and Certification - Supplement D, (Liquid Penetrant).

(Application for copies should be addressed to the American Society for Non-Destructive Testing, 4153 Arlingate Plaza, Columbus, OH 43228.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 1141 - Substitute for Ocean Water.
D 3951 - Standard for Commercial Packaging.
D 3953 - Strapping, Flat Steel and Seals.
D 4675 - Selection and Use of Flat Strapping Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1915 Race Street, Philadelphia, PA 19103.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section IX, Qualification Standard for Welding and Brazing Procedures, Welders, Brazers, and Welding and Brazing Operators.

(Application for copies should be addressed to the American Society of Mechanical Engineers, United Engineering Center, 354 East 47th Street, New York, NY 10017.)

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AMERICAN WELDING SOCIETY (AWS)

A5.1 - Covered Carbon Steel Arc Welding Electrodes.
A5.9 - Corrosion Resisting Chromium and Chromium-Nickel Steel Bare and Composite Metal Cored and Stranded Welding Electrodes and Welding Rods.
A5.10 - Bare Aluminum and Aluminum-Alloy Electrodes and Rods.
A5.18 - Steel, Carbon, Filler Metals for Gas Shielded Arc Welding.
A5.28 - Low-Alloy Steel Filler Metals for Gas Shielded Arc Welding.
D1.1 - Structural Welding Code - Steel.
D1.2 - Structural Welding Code - Aluminum.

(Application for copies should be addressed to the American Welding Society, 550 North LeJeune Road, P.O. Box 351040, Miami, FL 33126.)

ASSOCIATION OF AMERICAN RAILROADS (AAR)

Rules Governing Loading of Commodities on Open Top Cars.

(Application for copies should be addressed to the Association of American Railroads, 59 East Van Buren Street, Chicago, IL 60605.)

INTERNATIONAL STANDARDS ORGANIZATION (ISO)

ISO 1496-1 - Series 1 Freight Containers - Specification and Testing - Part 1: General Cargo Containers for General Purposes.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

NATIONAL FIRE PROTECTION AGENCY (NFPA)

National Electrical Code - NFPA 70, Article 310.

(Application for copies should be addressed to the National Fire Protection Agency, Batterymarch Park, Quincy, MA 02269.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

J185 - Access Systems for Off-Road Machines.
J429 - Mechanical and Material Requirements for Externally Threaded Fasteners.
J492 - Rivets and Riveting.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

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SOUTH COAST AIR QUALITY MANAGEMENT DISTRICT (SCAQMD)

Rule 442 - Usage of Solvents.

(Application for copies should be addressed to the South Coast Air Quality Management District, 9150 East Flair Drive, El Monte, CA 91731.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, (except for related associated detail specifications, specification sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The 3,000 GPH ROWPU shall be in accordance with Top Assembly Drawing TA13229E0000 and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviations from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to ensure correct fit, assembly, and operation of the ROWPU. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or the designated representative.

3.1.2 ISO container design requirements. The units shall meet the ISO 1496-1 requirements.

3.1.2.1 Certification. Each container shall be certified as required by the International Convention of Safe Containers and bear evidence of such certification (see 3.16.2 and 4.6.2).

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.4), in accordance with 4.3.

3.3 Material. Material shall be as specified herein and on the applicable drawings. Materials not specified shall be selected by the contractor and be subject to all provisions of this specification. The material shall have no adverse effect on the health of personnel when used for its intended purpose. The lines and all surfaces that come in contact with potable water shall conform to the Food, Drug, and Cosmetic, Food Additives (chapter 1, title 21,

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Code of Federal Regulations), for contact with food. The substances listed in sections 177.1020, 177.1040, 177.1050, and 177.1480 of the Food, Drug, and Cosmetic, Indirect Food Additives (chapter I, title 21, Code of Federal Regulations) shall not be acceptable materials. Questions pertinent to this effort shall be referred by the contracting officer to the appropriate departmental medical service who shall act as an advisor to the contracting agency (see 6.6).

3.3.1 Material deterioration prevention and control. The ROWPUs shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the item may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.3.1.3 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the ROWPU may be newly fabricated from recovered materials to the maximum extent practicable, provided the ROWPUs produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the ROWPUs.

3.3.2 Piping. The piping shall be color coded and painted or stenciled in accordance with MIL-T-704 and MIL-C-22750. The color coding shall be according to function as follows:

Function	Color
Raw water	Black
Backwash waste	Red
Filtered water	Yellow
Product water	Blue
Brine water	Purple

3.4 Operation. The ROWPU shall operate at design capacity without leaks, malfunction, or permanent deformation. The unit, when operated in accordance with the applicable technical manual, shall have a potable water production capacity of 3,000 GPH (60,000 GPD) on fresh or brackish water. The production rate for seawater shall be 2,000 GPH (40,000 GPD) (see 6.7.c) with all rates

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corrected to 77 °F by using the appropriate element manufacturer normalization chart. The ROWPU shall reduce water having a maximum TDS (total dissolved solids) of 37,000 mg/L to 1,500 mg/L or below. The system must be capable of treating a water source having maximum turbidity of 150 NTU (Nephelometric Turbidity Units). The product water produced by the unit shall meet the field supplies, long term treated water quality standards of chapter 3, TB MED 577, for physical, chemical, and bacteriological constituents except for the maximum permissible concentrations (mpc) of arsenic and cyanide. The mpc is 0.05 mg/L for arsenic and 0.5 mg/L for cyanide.

3.5 Treatment and painting. Treatment and painting shall be in accordance with MIL-T-704, or the applicable drawing unless otherwise noted. The painting shall be in accordance with MIL-C-53072 whenever a chemical agent resistant coating (CARC) is applied. Color shall be in accordance with FED-STD-595, or the applicable drawing. The exterior of the unit shall be painted green 34094 or tan 33448 (see 6.2). Chemical agent resistant paints shall be in accordance with MIL-C-46168 or MIL-C-53039. Untinted white clear or aluminized 17925 epoxy polyamide paint in accordance with MIL-C-22750 shall be used on the interior portions of the ISO container. MIL-P-53084 shall be used with CARC paints, MIL-C-46168, MIL-C-53039, or MIL-C-22750 as an alternative to epoxy primers, MIL-P-53022 or MIL-P-53030 when an electrodeposition painting process is used.

3.6 Environmental.

3.6.1 Operating temperature. When tested as specified in 4.5.1.3, the ROWPU shall operate in ambient air temperatures ranging from -25 to +110 °F. Special environmental equipment shall be supplied as required.

3.6.2 Storage temperature. When tested as specified in 4.5.1.3, the ROWPU shall not be damaged by storage at ambient temperatures from -28 to +145 °F. The reverse osmosis elements may be removed from the unit, if required, for storage at a controlled temperature.

3.7 Safety. The ROWPU shall be designed to minimize the probability and severity of injury to personnel. As a minimum the ROWPU shall conform to title 29, Code of Federal Regulations, chapter XVII, part 1910; and MIL-STD-454, requirement 1; and the following:

a. All reciprocating, rotating, or moving parts of the equipment shall be guarded when such parts are exposed to contact by personnel or otherwise create a hazard.
b. All points requiring lubrication during operation shall have fittings located or guarded so as to be accessible without hazardous exposure to personnel performing these operations.
c. Exposed electrical connections shall be guarded to protect personnel from hazardous currents and voltages as specified in MIL-STD-454. Guards or enclosures shall be made of nonconductive materials and be marked to indicate voltages present. The operators control panel shall be interlocked in accordance with MIL-STD-454, requirement 1, to

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de-energize all electrical components if opened. An interlock bypass switch shall be located inside the cabinet for servicing purpose.

d. Nonfunctional sharp edges, projecting points, and excessively long fastening devices shall be avoided.

e. Access to the chemical feeder(s) shall not cause the chemicals to splash.

f. Each container of chemicals shall be labeled (see 3.8) with the generic name and the following note:

"CAUTION: Protect eyes and skin from direct contact with chemicals. Be especially careful when opening containers to avoid inhaling chemical powders. Wash hands thoroughly with soap and water after handling chemicals."

Each chemical shall be provided with a Material Safety Data Sheet in accordance with FED-STD-313, (see 6.9). All containers shall be labeled with graduated markings indicating levels (see 3.8). Labels shall be on identification plates conforming to MIL-P-19834, type 1, style 2.

g. The discharge from the pressure-relief device for the high-pressure (h.p.) pump shall be controlled in a manner that shall prevent injury to personnel and damage to equipment.

h. Components subject to high operational temperatures (above 140 °F) and so located as to become a hazard to operating/maintenance personnel shall be guarded or insulated.

i. The green insulated equipment ground wire (EGW) shall not be connected to the ground return (neutral) at any point other than the generator or the secondary of a power distribution transformer. In all electrical equipment the AC power neutral (white wire) shall be isolated from the EGW by a minimum of 1 mega ohm DC resistance measured at the input connector. All electrical convenience outlets, raw water and distribution circuits shall have ground fault circuit interrupter (GFCI) protection or equivalent level of protection.

3.8 Human factors engineering. The ROWPU shall conform to accepted human factors engineering design criteria as described in MIL-STD-1472. Special design emphasis shall be given, but not limited to paragraphs 4 (General Requirements), 5.1 (Control/Display Integration), 5.2 (Visual Displays), 5.3 (Audio Displays), 5.4 (Controls), 5.5 (Labeling), 5.6 (Anthropometry), 5.7 (Ground Workspace Design Requirements), 5.8 (Environment), 5.9 (Design for Maintainer), 5.13 (Hazards and Safety) of MIL-STD-1472, as applicable.

3.9 Noise limits. When tested in accordance with 4.5.1.5, the noise produced by the ROWPU shall conform to MIL-STD-1474. The provisions of MIL-STD-1474, 4.3 and 4.4 shall be provided if and only if procedures in MIL-STD-1474 (5.1.1.2) have been pursued and documented to the satisfaction of the contracting activity and written permission to exceed the 82 dB(A) limit is obtained from the contracting activity. However, in no event will

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permission to exceed an 85 dB(A) noise limit at the ROWPU control station be granted. Hazard signs shall conform with MIL-STD-1474, 4.3 and will be readable at the distance indicated on the sign.

3.10 Reliability. When tested in accordance with 4.5.1.7, (see 6.7.f), the specified mean-time-between-failure (MTBF) of the ROWPU, not including the generator set, shall be 400 hours.

3.11 Maintainability.

3.11.1 Ease of maintenance. In accordance with 3.8, all major assemblies and installed attachments shall be accessible for maintenance, repair, and replacement without the removal of other major assemblies and installed attachments not normally removed. Covers or plates which must be removed for component adjustment, repair, replacement, or maintenance shall be equipped with quick-disconnect fasteners. Drain outlets shall be located for accessibility. Provisions shall be made to permit the use of receptacles for collecting drainage.

3.11.2 Maintenance support. Assemblages or support elements such as technical manuals, repair parts, special tools, or lubrication charts as specified, shall be adequate to perform the intended function of assisting or conducting maintenance operations.

3.11.2.1 Mean-time-to-repair (MTTR). The water purification unit shall demonstrate the following values for MTTR.

3.11.2.1.1 Unit level MTTR. The ROWPU shall exhibit a MTTR without the generator set of 1 hour or less at crew/unit level of maintenance.

3.11.2.1.2 Intermediate DS/intermediate GS MTTR. The ROWPU shall exhibit a MTTR without the generator set of 3 hours IDS/IGS (Intermediate Direct Support/Intermediate General Support) level of maintenance.

3.11.2.2 Maintenance ratio. The ROWPU shall have a maintenance ratio (see 6.7.f) of not more than 0.03 without the generator set.

3.11.2.3 Minimum-allowable-time-between scheduled maintenance. The minimum allowable time between scheduled maintenance excluding operator/crew checks and services shall be 250 hours.

3.12 Lubrication. When serviced with military lubricants (see 6.12), the ROWPU shall operate as specified herein.

3.12.1 Lubrication fittings. Lubrication fittings shall conform to MIL-F-3541, types I, II, and III. Fittings shall be located in a protected position and shall be accessible to a hand-operated grease gun conforming to MIL-G-3859 with a 10-inch flexible extension. The fittings shall be accessible without removing or adjusting parts. Panels and plates equipped

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with hand-operable quick disconnect fasteners may be removed to provide accessibility.

3.12.2 Pressure-release device. A pressure-release device shall be provided where the use of pressure lubricating equipment could damage grease seals or other parts.

3.12.3 Grease lubrication. All grease lubrication, including lubrication of sealed bearings, shall be with grease as described in MIL-G-23827. The ROWPU shall be assembled, run-in (broken-in), tested, and delivered with these greases. The bearings shall be cleaned before lubrication. A tag shall be attached in a conspicuous place to indicate which military grease has been used.

3.12.4 Oil lubrication. Upon completion of all testing, all components, including all engines, shall be drained and filled to the operating level with military oils under MIL-L-2104, MIL-L-21260, or MIL-L-46167 as appropriate. A tag or tags shall be attached in conspicuous places to indicate the temperature range, and the lubricant grade used.

3.12.5 Filling, draining, and checking provisions. Enclosures such as gear cases and transmission housings that contain a reservoir of lubricants for the lubrication of the enclosed parts shall be equipped with dipsticks, check plugs, or sight gauges to determine the level of the lubricant. The drain outlet shall be fitted with a magnetic drain plug conforming to MS35844 or MS49006. The drain outlet shall be located so that removal of the plug shall result in complete drainage of the lubricant from the enclosure. Drainage shall be from the drain outlet into containers on the ground when the ROWPU is in its normal position, and not onto other components. Integral tubes or trough shall be used to convey the lubricant from the drain outlet to the ground container.

3.13 Electromagnetic interference/electromagnetic pulse.

3.13.1 Electromagnetic interference. The electromagnetic interference emission characteristics of the ROWPU shall conform to MIL-STD-461.

3.13.2 Electromagnetic pulse (nuclear survivability) test. The ROWPU shall conform to the electromagnetic pulse (EMP) survivability criteria in (C) QSTAG 244 Edition 3 (U).

3.14 Fungus and moisture resistance. The electrical circuitry, including all components and connections, except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173, with 1 percent copper 8-quinolinolate (by weight) based on the nonvolatile content of the varnish. Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.

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3.14.1 Varnish. The varnish shall be applied by spray, brush, or a combination of both to give a dry-film thickness of not less than 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish. Components to be varnish coated shall be selected, sized, and installed such that their temperature ratings shall not be exceeded because of the thermal insulating effect of the coating.

3.14.2 Composition. Composition I shall be used, unless composition II is required, in order to comply with local air pollution regulations governing the application of the varnish. When composition II is used, the contractor shall provide evidence to the Government that the use of composition II is required and shall certify that the composition II material complies with Rule 442, South Coast Air Quality Management District.

3.15 Identification markings. The ROWPU shall be identified in accordance with MIL-STD-130. The marking shall be applied to the ROWPU plates conforming to MIL-P-514, type III, composition C, type I or II, composition A, class 1 material. Information included shall be as specified on the appropriate drawing. The plates shall be attached by rivets in a location where the plate shall be both visible and legible.

3.16 Data plates.

3.16.1 Shipping data plates. A shipping data plate conforming to MIL-P-514, type II, composition C, of type I, grade A, class 1 material shall be provided. In addition, the plate shall show the name, model number, shipping weight, gross vehicle weight, date shipped, and manufacturer. The plates shall be attached by rivets in a location where the plates shall be both visible and legible.

3.16.2 CSC plate. Provision shall be made to obtain the International Convention for Safe Containers (CSC) approval (see 3.1.2.1) in accordance with title 49 of the CFR. A CSC plate conforming to MIL-P-514, type II, composition C, of type I, grade A, class 1 material shall be provided. Space shall be provided for retest dates.

3.17 Plastic encapsulated information. ROWPU installation, maintenance, start-up, operation, and systems information shall be assembled into sets and shall contain information pertinent to the water treatment unit, and shall be stored in a metal pocket on the inside of the unit instrument and control panel. A plate stating "DO NOT REMOVE DRAWINGS FROM UNIT" shall be installed on the metal pocket. Each set shall be assembled using a locking ring which allows "flip through" of the set to the desired information without losing the other information sheets and shall be permanently attached in the area of the pocket by a non-metallic strap of four feet in length. The information to be provided need not be encapsulated until prior to final acceptance. However, all information to be provided shall be available at the time the first article testing starts. Drawings included as part of the required information shall be photographic (black line) reproduction of the original full size

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drawings. Duplicates of all information that has been encapsulated shall be included in the ROWPU operating manual. Plastic encapsulated information shall include but is not limited to the following:

a. Operation instructions. Plastic encapsulated operating instructions shall be complete but brief. The operating instructions shall describe procedures for starting, operating, and shutting down the unit at ambient temperatures from +110 to -25 °F. The operating instructions shall describe procedures for set-up, adjustment and operation of the unit instrument and control system.
b. Schematic and wiring diagrams. Plastic encapsulated schematic and wiring diagrams shall show the complete operational and functional sequence of both AC and DC unit circuitry for troubleshooting analysis and maintenance of the unit. Wiring diagram shall show the physical location of all wiring interconnections in the same relationship in which they are installed. Identification marking of the instruments, control devices and connections shall be shown in both diagrams and shall coincide with the markings on all items being identified. Electric symbols shall be in accordance with ANSI/IEEE 315. When additional electrical symbols are used for items not in ANSI/IEEE 315, the meaning of the symbols shall be defined by a legend on the diagrams.
c. Piping diagrams. Plastic encapsulated, color coded schematic diagrams shall show all ROWPU piping. These diagrams shall be color coded for identifying the various systems: clarification system, pressure tubing/membrane cartridges, membranes, backwash, piping, valves, control devices, gauges, tanks, pumps, and other components in the same physical relationships as they are installed.
d. System diagrams. Individual plastic encapsulated system diagrams shall be sufficiently detailed to permit system troubleshooting; shall show all inter-system components, controls, piping and instrumentation; shall show all inter-system interface connections and include instructions, settings and other data for operation, adjustment and troubleshooting of system components. Individual system diagrams shall be provided for the clarification system, reverse osmosis system, the backwash system, the chemical feed and membrane sterilization system.
e. Shipping preparation instructions. Plastic encapsulated instructions for blocking and bracing of components subject to in-transit damage, for preservation of unit systems and for securing of meters, relays, and controllers, etc., shall be provided. Complete instructions shall be provided for draining the unit to prevent freeze damage during transportation or storage.
f. Index of plastic encapsulated information. An index of the plastic encapsulated information shall be provided. The index shall contain a listing of all information and a cross reference to the ROWPU manual citing, as appropriate, the manual volume and section or chapter where the same information can be found. The index shall be permanently mounted on the unit interior wall near the control panel.

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3.18 Lifting and tiedown attachments diagram. A plate conforming to MIL-P-514, mounted in a conspicuous location on the exterior of the unit, shall have one or more diagrams showing the lift areas, the sling lifting attachments, slings, spreader facilities, and tiedown attachments for the unit.

3.19 Transportability. The ROWPU, with its components, shall be capable of being transported by military trailers, trains, marine vessels, and aircraft and must be capable of withstanding the impact forces encountered in shipment, without damage or permanent deformation. The ROWPU shall be equipped with tiedown or slinging provisions.

3.19.1.1 Air. The ROWPU shall conform to MIL-STD-1791 for air transportability requirements. The ROWPU, operator accessory pack, generator, and M871A1 trailer shall fit in no more than two C-130 or no more than one C-141 and C-5A aircraft. No components shall be damaged while the unit is being air transported. When required (see 6.2), the contractor shall provide sufficient data to the Government (see 6.6 and 6.14), to confirm that the ROWPU and components being supplied can be loaded and transported in a C-130, C-141 and C-5A.

3.19.1.2 Rail. The dimensional profile of the ROWPU shall be within the outline of the Gabarit International de Chargement (GIC) diagram shown in accordance with the AAR outline diagram. The container, with all associated ROWPU components in their transport configurations, shall withstand the shock resulting from the rail impact test in accordance with 4.5.1.10.2 without failure, damage or permanent deformation.

3.19.1.3 Highway. The ROWPU, when loaded on a semitrailer/tractor, shall be within the highway permit limits of all states.

3.19.1.4 Marine. The ROWPU shall be transportable by LARC-V and larger vessels operating over any type of waterway. The water treatment unit shall withstand without damage, the shock, rolling and pitching normally experienced in marine transportation on the deck or in the hold of a cargo vessel. For design purposes, the following conditions are assumed for marine transportation:

a. Normal roll shall not exceed 15 degrees.
b. Normal pitch shall not exceed 10 degrees.
c. Stacking the containers six high, when each is loaded to its rated gross weight of 17,500 pounds in accordance with ISO 1496-1.

3.19.1.5 Handling operation. The ROWPUs, complete with accessories, shall withstand vertical accelerations of 2.0g ±0.5g either upward or downward, during cargo terminal handling operations. The unit shall withstand these accelerations during crane lifting via its lifting slings and during lifting devices which employ the bottom-edge-pickup method.

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3.19.1.6 Lifting and tiedown attachments. The ROWPU shall have lifting and tiedown attachments, including slings and spreader bars, conforming to MIL-STD-209, type II, and IV as applicable. The maximum shipping weight for determining conformance to MIL-STD-209 shall be the weight of the ROWPU, accessories, and spare parts when properly prepared for transport. Lifting and tiedown attachments shall be labeled "LIFT HERE" and "TIEDOWN," respectively in 2-inch high letters in accordance with 3.8. Lugs or lifting eyes shall be mounted and arranged with the vertical center of gravity for safe handling and transport by overhead crane. The equipment when suspended shall hold perpendicular within 5 degrees of true vertical. The arrangement shall provide a stable and balanced lift with a load safety factor of at least five times the total weight of the equipment.

3.19.1.7 ROWPU forklifting provisions. The ROWPU shall be equipped with box section tine guide openings in the base for safe handling and transport by forklift truck. The box sections shall extend through the entire width of the equipment completely enclosing the forklift tines and shall be fabricated from metal having a minimum thickness of 1/4-inch. The box sections shall have cross-sectional dimensions of not less than 5 inches by 12 inches and shall be welded to the base. The tine guides shall be spaced not more than 48 inches apart center-to-center, and so spaced that a forklift truck, while transporting the equipment, can pass over pavement obstructions without causing the box sections to fracture or part from the base. Forklift box sections shall be located symmetrically with respect to center of gravity of the ROWPU. The forklift tine guide openings shall be labeled "FORKLIFT HERE" in .75-inch high letters above each opening in accordance with 3.8.

3.19.2 Mobility. The mobility of the system (trailer, ROWPU, generator set, and operator accessory pack) shall be no less than the stated off-road mobility for the M932 prime mover that is used. The total system (trailer, ROWPU, generator set, winterization kit, and operating accessory pack) shall not weigh more than 37,960 pounds.

3.20 Government-furnished property (GFP).

3.20.1 GFP for first article testing. The following property, in the quantities shown, shall be furnished by the Government (see 6.2) for first article testing:

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ITEM NO	DESCRIPTION	NSN	QUANTITY FOR EACH ROWPU
1	Tank, Lightweight Self-supporting	5430-01-170-6984	3
2	Semi-trailer 22-1/2 ton M871A2	2330-01-294-3367 or equivalent	1
3	60 KW Generator Set	6115-00-118-1243 or equivalent	1
4	100 Hr Overpack for 60 KW Generator:		
	Belts, V, Matched Set	3030-00-257-0727	1
	Gasket	5330-00-663-4773	3
	Filter, Element, Fluid	2910-00-287-1912	3
	Belt, V	3030-00-630-0092	1
	Gasket	5330-00-740-6097	2
	Element, Filter Fluid	2940-00-580-6283	2

3.20.2 GFP for production units. The following property, in the quantities shown, shall be furnished by the Government (see 6.2) for the production units:

ITEM NO	DESCRIPTION	NSN	QUANTITY FOR EACH ROWPU
1	Semi-trailer 22-1/2 ton M871A2	2330-01-294-3367 or equivalent	1

3.21 Workmanship. All parts, components, and assemblies of the system, including castings, forgings, molded parts, stampings, bearings, seals, machined surfaces and welded parts, shall be clean and free from sand, dirt, fine, pits, scale, flux, and other harmful extraneous material. All edges shall be rounded or chamfered in accordance with 3.8. The system shall be free from such defects as misaligned components, incomplete welds, rust, extraneous material, cracks, leaks, and other defects that could impair the operation and serviceability of the system.

3.21.1 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that shall not cause degradation of the original material characteristics. Corners shall be square and true. Flame-cutting, using tips suitable for the thickness of the steel, may be employed instead of shearing and sawing. All bends shall be made with controlled means to insure uniformity of size and shape. Precaution shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly. External surfaces shall be free of burrs, sharp edges

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and corners, except when sharp edges of corners are required or where they are not detrimental to safety.

3.21.2 Bolted connections. Bolt holes shall be accurately formed and shall have the burrs removed. Washers or lockwashers shall be provided where necessary. Matching thread areas securing bolts conforming to SAE J429 or capscrews shall be of sufficient strength to withstand the tensile strength of the bolt. All fasteners shall be correctly torqued and shall have full thread engagement.

3.21.3 Riveted connections. Rivets shall fill the holes completely. All rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member and shall be in accordance with SAE J492.

3.21.4 Welders, weldments, and welding.

3.21.4.1 Welders. Before assigning any welder or welding operator to welding work covered by this document, the contractor shall obtain certification that the welder or welding operator has passed qualification tests as prescribed by either of the following listed codes for the materials joined and the type of welding operation to be performed and that such qualification is effective as defined by the following codes: AWS D1.1, AWS D1.2 and ASME Boiler and Pressure Vessel Code, Section IX, Qualification Standard for Welding and Brazing Procedures.

Contractors who only make horizontal welds need not qualify welders for "all position welding". In the event of evidence of poor welds, the Government reserves the right to require retesting of any welder or welding operator.

3.21.4.2 Weld filler metals. Filler metal shall be used in all welds. Aluminum filler metal shall be in accordance with AWS A5.10. Steel filler metal shall be in accordance with AWS A5.1, A5.9, A5-18, A5.28 as appropriate. Low hydrogen filler electrodes shall be used for high-strength low alloy (HSLA) steel. In joints involving base metals of different groups, low hydrogen filler electrodes requirements applicable to the lower strength base metal shall be used. For corrosion-resisting steel, a low carbon filler electrode shall be used in accordance with AWS A5.9.

3.21.4.3 Aluminum and steel weldments. The surfaces and edges of parts to be welded including each layer of weld materials when multiple layers are required shall be free from oxide scale, paint, lubricant, and other foreign matter. Welds shall transmit stress to the heat-affected zone without permanent deformation or failure when the parts connected by the welds are subjected to proof and service loading. All welded parts shall be free from cracks, porosities, inclusions and other imperfections that may reduce the effectiveness of the part. Tack welds shall be subjected to the same quality requirements as final welds.

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3.21.4.3.1 Aluminum welding. Complete penetration, with uniform root reinforcement, and complete fusion of the metal shall be obtained on all welds. Finished aluminum welds may be ground, filed, chipped, or wire-brushed, but shall not be peened. Preheating for welding is permissible provided the temperature does not exceed 200 °F for a total not to exceed 30 minutes. The temperature is to be measured 3 inches from the weld joint, and is to be measured by temperature sensitive stick or paints. No process requiring the use of welding flux shall be used. Work shall be positioned and jigged for flat position welding whenever practicable.

3.21.4.3.2 Steel welding. Welds involving corrosion-resistant steel shall be passivated. Preheat and post heat treatment may be used on other steels as needed. Work shall be positioned for flat position welding whenever practicable.

3.21.4.3.3 Welding visual inspection criteria. Visual inspection shall be done with a X5 magnifier. Both the root (if available) and face shall be inspected. In addition to all requirements, the following shall apply:

a. There shall be no cracks in the weldment, heat-affected zone or adjacent base metal.
b. Weldment shall be free from slag inclusion.
c. There shall be complete fusion between base metal and weld metal, and between adjacent passes of weld metal.
d. There shall be no incomplete penetration unless specified as a partial penetration weld.
e. For base metals 0.25-inch or less, there shall be no undercut. Pipe greater than 0.25-inch in thickness shall not exceed 1/32-inch in depth undercut. Undercut must have a width no less than twice the depth.
f. There shall be no arc strikes. (Cracks or blemishes caused by arc strikes shall be ground to a smooth contour and checked to ensure soundness).
g. For a groove weld, the weldment shall be at least flush with surface of the base metal, and the weld reinforcement shall not exceed the following:

Base metal thickness, inches	Reinforcement, max. inches
3/8	3/32
over 3/8 to 3/4 inclusive	1/8
over 3/4	3/16

The contour of a fillet weld shall conform to AWS D1.1. For all welds, the weldment shall blend smoothly with the base metal. Overlap and abrupt weld face ridges are not permitted. Icicles are not permitted.

3.21.4.3.4 Welding dye penetrant inspection. Dye penetrant inspection shall be performed on all weldments in pipe with a working pressure of 150 pound-force per square inch gauge (psig) or more. The penetrant process shall

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be type I or II, method C in accordance with MIL-STD-6866. Rejectable defects are cracks in the weldment, heat affected zone or adjacent passes of weld metal, lack of fusion between base metal and weldment or between adjacent passes of weld metal, and piping (or elongated) porosity. A penetrant procedure shall be submitted to the contracting officer for approval. A record of inspection shall be made available to the Government upon request. Inspection personnel shall be certified in accordance with ASNT SNT-TC-1A. The Government reserves the right to perform any inspection set forth in the specification to assure that weldments conform to the prescribed requirements.

3.21.4.4 Welding procedure. Prior to any welding, the contractor shall prepare and submit to the contracting officer for approval, a written welding procedure with qualification results, in accordance with one of the following codes as applicable: AWS D1.1, AWS D1.2 and ASME Boiler and Pressure Vessel Code, section IX.

Copies of the welding procedure(s) shall be available at workspace for the welders, welding operators, tackers, and authorized inspectors. This procedure shall include, but is not limited to, the factors listed in table I. Welding procedure changes affecting any of the factors in table I shall be submitted as a new procedure.

TABLE I. Welding procedures.

Essential Variable	Required Resubmission
1. Welding process	*
2. Manual or automatic	*
3. Joint design	Change in basic weld type (groove to fillet, etc). A change in groove type (V- to U-groove), or decrease in the included angle or root opening of a groove, or an increase in the root face of a groove.
4. Base metal group or P-number	*
5. Base metal thickness range	*
6. Shielding gas composition	*
7. Shielding gas flow rate	Increase of 25% or a decrease of 10% in flow rate.

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TABLE I. Welding procedures - continued

Essential Variable	Required Resubmission
8. Filler metal electrode specification of F-number	Any change in specification of F-number, plus a change increasing the filler metal strength level (from E 70S to E 80S, for example).
9. Filler metal electrode size for each pass	A change in nominal diameter of more than 1/64-inch.
10. Single or multiple electrodes (sequence)	*
11. Use of backing material, consumable inserts or bridges	*
12. Weld position	*
13. In vertical welding, vertical up or vertical down for each pass	*
14. Preheat and interpass temperature	An increase in preheat temperature of 100 °F or more or decrease of 25 °F below minimum preheat or interpass temperature from approved condition.
15. Pre-welding and interpass cleaning	*
16. Voltage for each diameter electrode	A change of more than 7% of specified mean arc voltage for each diameter of electrode from approved condition.
17. Type of current AC or DC (including polarity)	*
18. Amperage for each diameter of electrode	A change of more than 10% of the specified mean arc amperage for each diameter of electrode from approved condition.
19. Travel speed (machine only)	A change of more than 25% of the number of passes specified for a specified groove from approved condition.

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TABLE I. Welding procedures - continued

Essential Variable	Required Resubmission
20. Single or multiple passes	A change of more than 25% of the number of passes specified for a specified groove from approved condition.
21. Machining of the weld	*
22. Post heat treatment	*
23. Inspection method (visual, PT, etc)	*
24. Method of defect removal (grinding, chipping, etc)	*
25. Method of inspection before rewelded	*
26. Repair welding process - If the same process and parameters as the original weld, the parameters used shall be listed	*
27. Code by which the welders or welding operators is to be qualified	*

* Any change

3.21.4.5 Rewelding defects. Portions of welded joints that have been rejected on inspection because of defects specified in 3.21.4.3 or not conforming to the weldment codes in 3.21.4.4 may be repaired only by rewelding. The defective weld shall be removed by chipping or machining. With the exception of steel, flame cutting shall not be used. Before rewelding, the joint shall be inspected to assure that all of the defective weld has been removed and that the joint is accessible so that the welding operator can obtain full penetration and fusion through the joint. Repairs in the form of patching aluminum shall not be performed.

3.21.4.6 Spot and seam welding. Spot and seam welding shall be in accordance with MIL-W-6858.

3.21.4.7 Brazing. Brazing of steels, copper alloys, and nickel alloys shall conform to MIL-B-7883.

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3.21.5 Diagrams, charts, and plates. Diagrams, charts, and plates attached to components shall be applied to give a surface free of wrinkles, bubbles, or other defects that may cause the diagrams, charts, and plates to become loose or damaged.

3.21.6 Cleaning. The system shall be thoroughly cleaned of all cutting oil, residue of tests, grease, dirt, metal scrap, and other contaminants. Cleaning shall be accomplished in a manner that shall not leave a residue or otherwise make the system unsuitable for its intended use. Except where moisture is required for preservation during shipment, the system shall be completely drained before preparation for packaging.

3.21.7 Soldering. All soldering shall be in accordance with 5.1.1.21, 5.1.1.2.1, and 5.1.1.9.9 of MIL-STD-11991. All subassemblies shall meet the soldering requirements for the next higher assembly in which it is incorporated.

3.22 NBC contamination survivability. The ROWPU must be able to survive NBC (Nuclear, Biological, Chemical) contamination (see 6.15) to the quantitative criteria of (C) QSTAG 747 (U), and shall be operable in a NBC environment (see 4.5.1.11).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

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4.1.2 Component and material inspection. The contractor is responsible for ensuring that components and materials are manufactured, examined and tested in accordance with referenced specifications, standards, and drawings, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.8).

4.3 First article inspection.

4.3.1 First article examination. The first article shall consist of one or more ROWPUs and shall be examined as specified table II in the order shown. Presence of one or more defects shall be cause for rejection.

TABLE II. Examination schedule.

Defects		Requirement Paragraph
101.	Assembly not complete or not correct.	applicable dwgs
102.	Parts or components missing.	applicable dwgs
103.	Parts or components not as specified.	applicable dwgs
104.	Material not as specified.	3.3 and applicable dwgs
105.	Treatment and painting not as specified.	3.5 and applicable dwgs
106.	Workmanship not as specified.	3.21
107.	System not completely drained and cleaned before preparation for packaging.	3.12.5, 3.21.6
108.	Identification not as specified.	3.3.2, 3.15, 3.17, 3.18, applicable dwgs
109.	Plastic encapsulated information missing or not complete.	3.17
110.	Shipping data plates missing or incorrect.	applicable dwgs, and 3.16
111.	Instruction plates missing, not attached, as specified or illegible.	applicable dwgs, 3.16
112.	Dissimilar metals not in compliance with MIL-STD-889.	3.3.1.1
113.	Welding not as specified.	3.21.4, applicable dwgs
114.	Soldering not as specified.	3.21.7
115.	Materials are not resistant to corrosion or deterioration or treated to prevent corrosion or deterioration.	3.3.1, applicable dwgs

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TABLE II. Examination schedule - continued

Defects	Requirement Paragraph
116. Contractor does not have documentation available for identification of materials, material finishes, or treatments.	3.3.1.2
117. Lubricants not as specified.	3.12.3, 3.12.4
118. 100 hour overpack nonexistent or incorrect.	applicable dwgs
119. Military lubes not used.	3.12, applicable dwgs
120. Incorrect lube fittings.	3.12.1, applicable dwgs
121. No pressure-release device on lube areas.	3.12.2, applicable dwgs
122. Military greases not used and properly labeled.	3.12.3, applicable dwgs
123. Equipment not re-lubricated after testing.	3.12.4
124. Safety not as specified.	3.7, applicable dwgs
125. Chemical containers not labeled as specified.	3.7, applicable dwgs
126. Varnish non-existent or not in compliance with standards.	3.14, applicable dwgs
127. Identification plates incorrect or missing.	3.15, applicable dwgs
128. Instruments not identified properly.	3.15, applicable dwgs
129. Grounds not marked correctly.	3.15, applicable dwgs
130. Voltage not correctly marked at outlets.	3.15, applicable dwgs
131. Hazard plates incorrect or missing.	3.15, applicable dwgs
132. Shipping data plate incorrect or missing.	3.16.1, applicable dwgs
133. Dimensions not accordance with the air transportability requirements of MIL-STD-1791.	3.19.1.1
134. Welders and welding operators not certified as specified.	3.21.4.1
135. Subsystems and components not operating as required.	Applicable dwgs
136. Any part or component not in accordance with QAP requirements as shown on the drawings.	Applicable dwgs
137. Any dimension, other than those identified by the QAP requirements shown on the drawings, not as specified.	Applicable dwgs

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TABLE II. Examination schedule - continued

Defects	Requirement Paragraph
138. The system not completely drained and cleaned before preparation for packaging.	3.21.6, 3.21
139. Filling, draining, and checking provisions not as specified.	3.12.5

4.3.2 First article test. Upon successful completion of the examination specified in 4.3.1 and contractor start-up and performance demonstration test (50-hour test) specified in 4.7, the first article shall be subjected to the tests in table III. Failure of any test shall be cause for rejection.

TABLE III. Test schedule.

First Article	Characteristics	Requirement Paragraph	Test Para.
X	Operational test (1500 hrs)	3.4	4.5.1.2
X	Environmental	3.6	4.5.1.3
X	Safety	3.7	4.5.1.4
X	Human factors	3.8	4.5.1.6
X	Noise level	3.9	4.5.1.5
X	Reliability (MTBF)	3.10	4.5.1.7
X	Maintainability	3.11	4.5.1.8
X	Electromagnetic interference	3.13	4.5.1.9.1
X	Nuclear survivability	3.13.2	4.5.1.9.2
X	Transportability	3.19	4.5.1.10
X	CSC certification	3.1.2.1, 3.16.2	4.6
X	NBC contamination survivability	3.22	4.5.1.11

First article test failure shall be defined as any malfunction which causes or may cause degradation of equipment performance capability/requirements below specified levels; or serious personnel safety hazards. Simultaneous, related malfunctions shall be considered as one failure. Normal prescribed operator adjustments/maintenance shall be excluded, as will crew correctable maintenance actions completed in one hour or less using on-board tools and equipment. In addition, any test peculiar incident or failures resulting from causes other than the ROWPU itself shall not be considered a failure. The Government reserves the right to determine which incidents are failures. Failures under this definition shall be valid regardless of when they occur. For example, a failure occurring during maintenance or after a deliberate shutdown is still valid although the unit may not be operating.

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4.4 Quality conformance inspection.

4.4.1 Examination. Each ROWPU shall be examined for the characteristics listed in 4.3.1, table II. Presence of one or more defects shall be cause for rejection.

4.4.2 Test. Upon successful completion of the examination in 4.3.1, each ROWPU shall be subjected to the 6-hour operational test (see 4.4.2.1). Failure of any test shall be cause for rejection.

4.4.2.1 Quality assurance test (6 hours). Each ROWPU shall be hydraulically tested using the same feedwater specified in 4.7 and maintaining the proper operating pressure for not less than 6 water production hours. Failure to exhibit the minimum total acceptable product flow specified in 3.4 shall be a criterion for failure of this test. In addition, evidence of any one of the following conditions shall constitute failure of this test:

a. Leaks at any weld seam or any connection.
b. Any light or switch fails to operate.
c. Valves fail to operate properly.
d. Mechanical or electrical failure of any component or part.

Failure of the unit to exhibit a total dissolved solids rejection as specified in 3.4 shall not constitute criterion for failure of the 6-hour operational test. Upon completion of the test, the ROWPU shall be flushed with fresh chlorine-free water for a period of 10 minutes. Chemical tanks are to be filled with fresh water. Control knobs on the chemical feed pumps are to be set for maximum flow. The ROWPU is to operated in a normal manner for 10 minutes. After flushing, all drain and vent valves are to be opened and left open. The ROWPU shall then be prepared for movement as instructed by the technical manuals.

4.5 Inspection schedule.

4.5.1 Tests.

4.5.1.1 Test conditions. Unless otherwise specified in the technical manuals, tests shall be performed without shelter and at the environmental conditions existing at the place of test. The system shall operate as specified herein without maintenance other than the recommended normal scheduled maintenance as described in the technical manual(s). Prior to start of the test components of the system requiring lubrication shall be lubricated with oils and greases specified in 3.12. Oils shall be those designated for use in the ambient temperature at the place of test.

4.5.1.2 Operational test (first article testing). The 1,500 reliability operating hours shall be divided equally among the first article test units, with one-fourth of the test time done with seawater and the three-fourths with fresh water (see 6.7.a). Water quality and production rate data shall be collected each 2-hour period. Test unit(s), as specified, shall complete a

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250-mile (400 km [kilometers]) roadability test at an average speed of 20 mph (miles per hour) (32 km/hr [kilometers per hour]) over dirt, and dirt and gravel roads. The contractor shall provide a system support package consisting of the necessary supplies, chemicals, spares, repair parts and tools required for the necessary amount of testing as defined above on each first article water purification unit. One unit shall complete a 48-hour seawater spray test as specified in 4.5.1.3.1 and a 20-hour rain test as specified in 4.5.1.3.2. All the requirements of this specification are to be met. Normalization to 77 °F shall be done in accordance with the appropriate element manufacturer normalization chart (see 6.7 for definitions of terms).

4.5.1.3 Environmental test. When specified (see 6.2), any or all of the following tests shall be performed as part of the first article test.

4.5.1.3.1 Seawater spray test. The ROWPU shall be exposed to a synthetic seawater spray for 48 hours. The mist shall be produced by spraying the unit with a hose that has a low pressure nozzle. The nozzle pressure shall be as low as practicable to produce a fine spray. Its orifices shall be between 0.02- and 0.03-inch in diameter. At the end of exposure period, the unit shall be operated in accordance with the 6-hour quality assurance test (see 4.4.2.1); all the requirements of this specification are to be met, only failure relating to corrosion, water penetration, and water damage shall be assessed for this test.

4.5.1.3.2 Rain test. The ROWPU shall be exposed to a simulated rainfall of 2.0 +0.5 inch per hour for 10 hours. The simulated rainfall shall fall at an angle of not less than 60 degrees to the container's horizontal plane. After 10 hours, the unit shall be operated in accordance with the 6-hour quality assurance test (see 4.4.2.1), and shall be required to meet all the requirements of this specification. Only failures relating to corrosion, and water damage shall be assessed for this test. The rain operation cycle shall be repeated twice, for a total of 20 hours of exposure to rain and 12 hours of operation.

4.5.1.3.3 Cold environment test. The ROWPU shall be put into a 24-hour cold soak at -28 °F, and shall then be deployed and operated at -25 °F for 100 water production hours. During the 100 water production hours of operation, the unit shall be shut down and restarted at equal intervals four times after a two-hour wait (see 6.7.d). The waiting period shall not be included in the 100 hours of operation. All operation, shut downs and restarts shall be in accordance with the operations manual, and the ROWPU shall be required to meet all the requirements of this specification. Failures shall constitute nonconformance to this test.

4.5.1.3.4 High temperature test. The ROWPU shall be tested in accordance with MIL-STD-810, method 501.2, procedure I. The maximum test temperature for storage shall be 145 °F, and for operating shall be 110 °F. Operation shall be in accordance with the 6-hour operational test specified in 4.4.2.1. Failures shall constitute nonconformance of this test.

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4.5.1.4 Safety. Safety evaluation shall be accomplished throughout testing. Nonconformance to 3.7 shall constitute failure of this test.

4.5.1.5 Noise level test. Noise levels shall be measured in accordance with MIL-STD-1474 requirements and reported in the format indicated by MIL-STD-1474, figure 11. As a minimum, noise levels shall be measured when equipment is operating under full load. Nonconformance to 3.9 shall constitute failure of this test.

4.5.1.6 Human factors. The water purification units shall be inspected throughout testing to determine the human factor characteristics. Nonconformance to 3.8 shall constitute failure of this test.

4.5.1.7 Reliability. When specified, using the mean-time between failure (MTBF) specified in 3.10, the ROWPU shall be tested as specified in 4.5.1.1 and 4.5.1.2 for 1,500 reliability operating hours (see 6.7.e). If five or fewer failures occur, the unit(s) shall be considered acceptable.

4.5.1.8 Maintainability. Maintenance actions occurring during the operational test of 4.5.1.2 shall be documented and sufficient data shall be collected to determine compliance with 3.11 (i.e., man-hours, clock hours, maintenance level prescribed by the manuals and used parts replaced, comments on ease of maintenance and any other data directed by the Government). All preventive and corrective maintenance shall be performed according to the maintenance manuals. Nonconformance to 3.11 shall constitute failure of this test.

4.5.1.9 Electromagnetic interference/electromagnetic pulse test.

4.5.1.9.1 Electromagnetic interference. The ROWPU shall be tested in accordance with MIL-STD-461 and MIL-STD-462 to determine conformance to 3.13.1. The findings shall be reported on the generator and the ROWPU operating together, and on the generator operating along. Nonconformance to 3.13.1 shall constitute a failure of this test, except that failure attributable solely to GFP shall not be charged against the ROWPU.

4.5.1.9.2 Electromagnetic pulse (nuclear survivability) test. The ROWPU shall be tested to determine conformance to 3.13.2. The findings shall be reported on the generator and the ROWPU operating together, and on the generator operating along. Nonconformance to 3.13.2 shall constitute a failure of this test except the failures attributable solely to GFP shall not be charged against the ROWPU.

4.5.1.10 Transportability.

4.5.1.10.1 Air transport test. The ROWPU shall be tested in accordance with MIL-STD-1791 to provide conformance to 3.19.1.1. Nonconformance to 3.19.1.1 shall constitute failure of this test.

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4.5.1.10.2 Rail impact test. The first article ROWPU shall be tested in accordance with MIL-STD-810 to provide conformance to 3.19.1.2. Nonconformance to 3.19.1.2 shall constitute failure of this test.

4.5.1.10.3 Highway transport test. The ROWPU shall be tested as specified in 3.19.1.3. Nonconformance to 3.19.1.3 shall constitute failure of this test.

4.5.1.10.4 Marine transport test. The ROWPU shall be tested as specified in 3.19.1.4. Nonconformance to 3.19.1.4 shall constitute failure of this test.

4.5.1.10.5 Lifting and tiedown test. The lifting and tiedown provisions shall be tested in accordance with MIL-STD-209 to prove conformance to 3.19.1.6. Nonconformance to 3.19.1.6 shall constitute failure of this test.

4.5.1.11 NBC contamination survivability test. The ROWPU shall be exposed to simulated or actual NBC agents and then decontaminated in accordance with the operations manual to determine conformance with 3.22. After each decontamination the unit shall be operated in accordance with the 6-hour quality assurance test (see 4.4.2.1), shall be required to meet all the requirements of this specification. Failures, as defined, shall constitute nonconformance to this test. The NBC contamination-decontamination cycle shall be repeated five times.

4.6 ISO container tests. The contractor shall conduct the following tests and submit the results for Government approval on at least one of the ROWPU ISO containers prior to the start of first article testing:

a. Stacking. The unit shall be subjected to the stacking tests of ISO 1496-1.
b. Lifting from the four top corner fittings. The unit shall be subjected to the lifting from the four corner fittings tests of ISO 1496-1.
c. Lifting from the four bottom corner fittings. The unit shall be subjected to the lifting from the four bottom corner fittings tests of ISO 1496-1.
d. Restraint. The unit shall be subjected to the restraint tests of ISO 1496-1.
e. Strength of end walls. The unit shall be subjected to the strength of end walls tests of ISO 1496-1.
f. Strength of side walls. The unit shall be subjected to the strength of side walls tests of ISO 1496-1.
g. Strength of the roof. The unit shall be subjected to the strength of the roof tests of ISO 1496-1.
h. Transverse rigidity. The unit shall be subjected to the transverse rigidity tests of ISO 1496-1.
i. Longitudinal rigidity. The unit shall be subjected to the longitudinal rigidity tests of ISO 1496-1.

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j. Lifting from the fork lift pockets. The unit shall be subjected to lifting from the fork lift pockets tests of ISO 1496-1.
k. Floor loading. The unit shall be tested for floor loadings in accordance with ISO 1496-1.

4.6.1 Inspection failure. Failure of a first article water purification unit to meet any requirements specified herein, during, and as a result of the examination and tests specified in 4.3 and 4.5 shall be cause for rejection of the first article.

4.6.2 Additional tests. International Convention of Safe Container approval shall be obtained for any additional tests required to receive certification.

4.7 Contractor start-up and performance demonstration test (50-hour). Each of the first article water purification units shall be operated on seawater not less than 50 water production hours with at least 2 back washing cycles during the 50-hour period and with a 50-mile (80 Km) roadability test on dirt or gravel roads at an average speed of 20 mph between the first 25 hours and the second 25 hours. Salt conforming to ASTM D 1141, (substitute ocean water, formula A, section 6, without heavy metals) shall be used to make up the test feedwater. The salt shall be dissolved in dechlorinated tap water to provide a solution containing 19,000 ±1,000 mg/l, chloride ion. Sodium thiosulfate may be used for removal of chlorine. The permeate and concentrate streams may be blended for recirculation provided that the temperature of the feed does not exceed 90 °F. Normalization to 77 °F shall be done in accordance with the appropriate element manufacturer normalization chart. The water purification unit shall be set up, operated, and maintained in accordance with the applicable technical manual. The ROWPU must meet the requirements specified in this specification before the units may be shipped to first article testing. In addition, evidence of any one of the following conditions shall constitute failure of this test:

a. Leaks at any weld seam or any connection.
b. Light, or any switch fails to operate.
c. Valves fail to operate properly.
d. Mechanical or electrical failure of any component or part.
e. Failure of the unit to exhibit a total dissolved rejection as specified in 3.4.
f. Failure of unit to exhibit the production rate specified in 3.4.

Upon completion of test, the units shall be flushed with fresh chlorine free water for a period of 10 minutes. The chemical utility tanks shall be filled with fresh water. All control knobs on the chemical feed pump are to be set to open. The unit shall be operated in a normal manner for the 10-minute period. After flushing, all vents and drain valves are to be opened, and left open.

4.8 Inspection of packaging.

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4.8.1 First article pack inspection. The first article pack shall be examined for the defects specified in 4.8.2.3. Presence of one or more defects shall be cause for rejection.

4.8.2. Quality conformance inspection of packaging.

4.8.2.1 Unit of product. For the purpose of inspection, a completed pack shall be considered a unit of product.

4.8.2.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.8.2.3 Examination. Samples selected in accordance with 4.8.2.2, shall be examined for the following defects. One or more defects shall be cause for rejection.

140. Reverse osmosis test elements or tubes not removed from the unit; the unit not drained and valves left open as specified for level A or C (see 5.2.1.1 or 5.2.2).
141. Caps not installed on water inlets and outlets as specified for level A or C (see 5.2.1.1 or 5.2.2).
142. Hardware items requiring a contact preservative not coated as specified for level A or C (see 5.2.1.2 or 5.2.2).
143. Ground rods not secured in the brackets as specified for level A or C (see 5.2.1.3 or 5.2.2).
144. Trailer chassis not preserved as specified for level A or C (see 5.2.1.4 or 5.2.2).
145. Raw water pumps, backwash pump, booster pump, and distribution pump not preserved as specified for level A or C (see 5.2.1.5 or 5.2.2).
146. Canvas hoses not coiled and secured as specified, and hard rubber hoses not left straight and secured on the unit as specified for level A or B (see 5.2.1.6 and 5.2.2).
147. Technical publications not preserved as specified for level A or C (see 5.2.1.7 or 5.2.2).
148. Hand tools not preserved as specified for level A or C (see 5.2.1.8 or 5.2.2).
149. Extension light cord not coiled or hanked and secured as specified for level A or C (see 5.2.1.9 or 5.2.2).
150. Adapters, nipples, bushings, nozzle and strainer not wrapped with barrier material as specified for level A or C (see 5.2.1.10, 5.2.1.11, or 5.2.2).
151. Box closure for the conductivity meter and gate valves not method V as specified for level A or C (see 5.2.1.12, 5.2.1.14, or 5.2.2).
152. Flashlight not preserved as specified for level A or C (see 5.2.1.13 or 5.2.2).
153. Reverse osmosis elements, cleaners, polyelectrolyte and sequestrant not preserved a specified for level A or C (see 5.2.1.15 or 5.2.2).
154. Box of lamps not preserved as specified for level A or C (see 5.2.1.16 or 5.2.2).

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155. Cargo straps not preserved as specified for level A or C (see 5.2.1.17 or 5.2.2).
156. Generator not preserved in accordance with the referenced document for level A or C (see 5.2.1.19 or 5.2.2).
157. The chemicals not preserved as specified for level A or C (see 5.2.1.20 or 5.2.2).
158. Components not specifically preserved, not preserved as specified for components of similar design or construction as specified for level A or C (see 5.2.1.21 or 5.2.2).
159. Consolidation of components, elements, chemicals and accessories not as specified for level A or C (see 5.2.1.22 or 5.2.2).
160. Boxed and unboxed components, accessories and support equipment not secured in their appropriate places inside the ISO container for level A or C (see 5.2.2 or 5.3.1).
161. Trailer mounted pump (h.p. pump skid) and if applicable, the generator, not fitted with a box type cover and the cover secured as specified for level A (see 5.3.1).
162. When applicable, the generator not packed, blocked and braced and strapped as specified for level A (see 5.3.1).
163. Marking missing, illegible, incorrect or incomplete for level A or C (see 5.4).

5. PACKAGING

5.1 First article pack. The contractor shall furnish a first article pack for examination within the time frame specified (see 6.2) to prove prior to starting production packaging that the applied preservation, packing, and marking comply with the packaging requirements of this specification. Examination shall be as specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.4). The first article pack may be prepared utilizing either the first article model or a production water purification unit. When the first article model is utilized, any preservation and packing shall be removed by the contractor at no expense to the Government, when requested by the Government to facilitate comparison between the first article model and production water purification units.

5.2 Preservation. Preservation shall be level A or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Test elements. Reverse osmosis elements used for test purposes shall be removed from the unit; the unit drained and the drain valves left open. Install caps on water inlets and outlets.

5.2.1.2 Hardware items. Hardware items such as bolts, nuts, linkages, pins, grease fittings, and capscrews requiring the application of a contact preservative in accordance with MIL-P-116 shall be coated with type P1 or P19 preservative. The preservative shall conform to the applicable specification listed and shall be applied in accordance with MIL-P-116.

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5.2.1.3 Ground rods. Ground rods shall be secured in the brackets provided on the unit.

5.2.1.4 Trailer chassis. The components of the trailer chassis shall be preserved in accordance with MIL-V-62038, level A (mobile).

5.2.1.5 Raw water pumps, backwash pump, booster pump and distribution pump. These pumps shall be preserved in accordance with MIL-P-10603, level A.

5.2.1.6 Hose assemblies. Canvas hoses shall be coiled in minimum size coil. The coils shall be secured with cotton tape or twine. Hard rubber hoses shall be left straight for consolidation on the unit.

5.2.1.7 Technical publications. Technical publications shall be preserved in accordance with MIL-P-116, method IC-1 or IC-3.

5.2.1.8 Hand tools. Hand tools shall be preserved in accordance with PPP-P-40, level A.

5.2.1.9 Extension light. The cord shall be coiled or hanked to occupy minimum space and shall be secured with cotton tape or twine.

5.2.1.10 Adapters, nipples and bushings. Each unassembled adapter, nipple, and bushing shall be preserved in accordance with MIL-P-116, method IC-1 or IC-3 or wrapped with barrier material conforming to PPP-B-1055, class B-2 or MIL-B-121, type I, grade A, class 2. The barrier material shall be secured with tape conforming to PPP-T-76.

5.2.1.11 Nozzle and strainer. The nozzle and strainer shall each be preserved in accordance with MIL-P-116, method IC-1 or IC-3 or wrapped with barrier material conforming to PPP-B-1055, class B-2, or MIL-B-121, type I, grade A, class 2. The barrier material shall be secured with tape conforming to PPP-T-76.

5.2.1.12 Conductivity meter. The conductivity meter shall be preserved in accordance with MIL-P-116, method 11b. The inner container shall be the vendor's carton or a box conforming to PPP-B-636, type CF, class domestic, variety SW, grade 125, style optional and the outer container shall be a box conforming to PPP-B-636, W6c, style optional. Box closure for the outer container shall be in accordance with the appendix to the box specification, method V.

5.2.1.13 Flashlight. The flashlight shall be preserved in accordance with MIL-P-116, method IC-2 using the vendor's carton or a box conforming to PPP-B-636, type CF, weather-resistant, variety SW, grade W6c, style optional.

5.2.1.14 Gate valves. Each gate valve shall be placed and secured in a close-fitting box conforming to PPP-B-636, W6c, style RSC. Box closure shall be in accordance with the appendix to the box specification, method V.

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5.2.1.15 Elements, cleaners, polyelectrolyte, and sequestrant. Elements, cleaners, polyelectrolyte, and sequestrant shall be placed into a snug-fitting box conforming to PPP-B-636, W6c. Box closure shall be in accordance with the appendix to the box specification, method V.

5.2.1.16 Box of lamps. The lamps shall be preserved in accordance with MIL-P-116, method IC-2, using the box provided as the container.

5.2.1.17 Cargo straps. Cargo straps not used for securing equipment in position shall be preserved in accordance with MIL-P-116, method IC-1 or IC-3.

5.2.1.18 Filter tubes. One set of filter tubes shall be installed in the cartridge filter upon completion of testing. The other set (8 each) shall be placed in the storage box on the unit.

5.2.1.19 Generator. The generator shall be preserved in accordance with the level A requirements of MIL-G-28554.

5.2.1.20 Chemicals. The chemicals shall be placed in a snug-fitting box conforming to PPP-B-636, weather-resistant. Box closure shall be in accordance with the appendix to the box specification, method V.

5.2.1.21 Other components. Other components not specifically mentioned herein requiring protection from corrosion or physical or mechanical damage shall be preserved as specified for components of similar design or construction.

5.2.1.22 Consolidation. All components and accessories intended for consolidation shall be as indicated in drawing 13229E1378.

5.2.1.22.1 Elements, cleaners, polyelectrolyte, sequestrant and chemicals. Elements, cleaners, polyelectrolyte, sequestrant and chemicals shall be consolidated into a box conforming to PPP-B-601, overseas type, style optional. Box closure shall be as specified by the appendix to the box specification and strapping shall be in accordance with ASTM D 3951, type 1 or 2, zinc-coated, size as applicable and ASTM D 4675.

5.2.2 Level C. Each purification unit, components accessories and support equipment shall be preserved as specified for level A except all boxes shall be domestic class or type, whichever is applicable and the generator shall be preserved in accordance with the level C requirements of MIL-G-28554.

5.3 Packing. Packing shall be level A or C as specified (see 6.2).

5.3.1 Level A. Each complete purification unit with all boxed and unboxed components accessories and support equipment shall be secured or strapped in its appropriate place inside the ISO container in accordance with drawing 13229E1378. The pump outside the ISO container shall be protected with a modified box cover of ends, sides and top constructed in accordance with

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PPP-B-601, overseas type, style I. The box cover shall be the minimum size required to fit over and completely enclose the pump structure above the trailer chassis; plywood shall be 3/8-inch thick. The cover shall not interfere with lifting or towing of the unit. The box cover shall be bolted in place. If space allows, the generator shall also be protected by the box cover specified herein. Otherwise, the generator shall be packed in a close-fitting box conforming to PPP-B-601, overseas type, style I. Blocking and bracing shall be in accordance with MIL-STD-1186. Strapping shall be as specified in 5.2.1.22.1. The generator shall be shipped separate from but at the same time as the purification unit.

5.3.1.1 Elements, cleaners, polyelectrolyte, sequestrant and chemicals. The elements, cleaners, polyelectrolyte, sequestrant and chemicals shall not require packing but shall be shipped separate but at the same time as the purification unit.

5.3.2 Level C. The complete purification unit shall be packed as specified for level A except the boxes shall be domestic type and strapping may be zinc-coated.

5.4 Marking. In addition to any special marking specified in the contract or purchase order (see 6.2), marking shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The water purification unit is intended to supply potable water from any fresh, brackish, or seawater source. The source may be contaminated with nuclear, biological, or chemical warfare agents.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When a first article is required for inspection (see 3.2 and 4.3).
d. When the Government will conduct any or all of the first article examination and tests, the contracting officer shall specify which examination and test will be conducted by the Government and which shall be conducted by the contractor (see 3.2).
e. Color of the interior unit (see 3.5).
f. When air transportability data is required (see 3.19.1.1, 6.5, and 6.14).
g. Government furnished property (see 3.20.1 and 3.20.2).
h. When environmental test is required (see 4.5.1.3).
i. When a transportability report is required (see 6.5).
j. Time frame required for submission of first article pack (see 5.1).

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k. Level of preservation and packing required (see 5.2. and 5.3).
l. Any special marking (see 5.4).

6.3 Additional tests. Additional tests within the scope of this specification may be directed by the contracting officer.

6.4 First article. When a first article inspection is required, the items should be a production model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.5 Transportability Report. When specified (see 6.2), the contracting officer should arrange for the contractor to furnish a transportability report in accordance with AR 70-47 and S.O.P 70-38 for the ROWPU, 95 days prior to delivery of the first article ROWPU.

6.6 First article pack. Any changes or deviations of production packs from the approved first article pack will be subject to the approval of the contracting officer. Approval of the first article pack will not relieve the contractor of his obligation to preserve, package, pack, and mark the water purification units in accordance with this specification.

6.7 Definitions.

a. Fresh water: Water with total dissolved solids (TDS) less than 1,500 mg/L.
b. Brackish water: Water with TDS greater than 1,500 mg/L and less than or equal to 15,000 mg/L.
c. Seawater: Water with TDS greater than 15,000 mg/L and less than or equal to 37,000 mg/L.
d. Water production rate calculation: (all rates corrected to 77 °F by using the appropriate element manufacturer normalization chart)

Average Hourly Water Production in Gallons Per Hour (GPH) = Total water produced (gal)/Water Production Time (clock hours)

Average Daily Water Production in Gallons Per Day (GPD) = Total water produced (gal)/Operating Time (clock hours)/24

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Where:

Water Production Time = elapsed clock time while product water is being produced.

Operating Time = water production time + off line PMCS (Preventitive Maintenance Crew Service) time + off line crew service time + unscheduled maintenance action time (nondeferrable) + set-up and teardown time.

e. Reliability operating time: Water production time + media filter backwash time + RO element cleaning time (recirculation time) + interior basket strainer cleaning time + cartridge changing time.
f. Maintainability: Maintenance ratio is defined as the ratio of the total active maintenance manhours required (scheduled and unscheduled) to the reliability operating time. Manhours for repair of replaced components are excluded if they are repaired at depot level maintenance. Manhours for crew services including backwashing, cleaning, and preventive checks and services are excluded from the maintenance ratio.

6.8 Medical advisor. The Surgeon General of the Army will act as advisor to make the necessary determination on health hazards. Inquiries should be forwarded to Headquarters, Department of the Army, ATTN: DASG-TSP-E, Washington, DC 20310.

6.9 Noise limits. When the noise level of the water purification unit exceeds 82 dB(A) (see 3.9), the appropriate discussion of the noise hazard in accordance with MIL-STD-1474 (MANUALS Paragraph) shall include: (a) requirement for hearing protection; (b) the noise level of the water purification unit, and (c) what operational situation or system configuration will reduce the level to 82 dB(A).

6.10 Technical manuals. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, military specifications and standards that have been cleared and listed in DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL) must be listed on a separate Contract Data Requirements List (DD Form 1423), which is included as an exhibit to the contract. The technical manuals must be acquired under separate contract line item in the contract.

6.11 Material Safety Data Sheets. Contracting officers shall identify those activities requiring copies of completed Material Safety Data Sheets prepared in accordance with FED-STD-313. The pertinent Government mailing addressed for submission of data are listed in appendix B of FED-STD-313.

6.12 Lubricants. MIL-STD-838 prescribes the policy for using specification-type products wherever possible and provides specific requirements for potential use of non-standard proprietary products.

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MIL-STD-838 is implemented by MIL-HDBK-113. The contracting officer should note that unless otherwise authorized by the US Army Belvoir Research, Development, and Engineering Center, ATTN: STRBE-VF, Fort Belvoir, VA 22026-5606, lubricants, fluids, and greases for ground equipment systems must be restricted to those listed under MIL-HDBK-113, chapter 2.

6.13 Government-furnished equipment. The contracting officer should arrange to furnish the property specified in 3.20.1 and 3.20.2.

6.14 Consideration of data requirements. The following data requirements should be considered when this document is applied on a contract. The applicable Data Item Description (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/ provided and that the DIDs are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference paragraph	DID Number	DID Title
a. 3.19.1.1	DI-PACK-80880	Transportability Report
b. 3.21.4.1	DI-H-24029A	Certification, Welding Qualification

The above DID was that cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

6.15 International standardization agreement. Certain provisions of this specification (see 3.13.2 and 3.22) are the subject of international standardization agreements (QSTAG 244 (C) and QSTAG 747 (C)). When amendment, revision, or cancellation of this specification is proposed that will modify the international agreement concerned, the preparing activity will take appropriate action through international standardization channels, including departmental standardization offices, to change the agreement or make other appropriate accommodations.

6.16 NBC contamination. NBC comtamination survivability shall be in accordance with AR 70-71 (see 3.22).

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6.17 Subject term (key word) listing.

Water, brackish
Water, contaminated
Water, fresh
Water, potable
Water, salt

Custodian:
Army - ME

Preparing activity:
Army - ME

Project 4610-A125

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

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RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-W-53122(ME)	2. DOCUMENT DATE (YYMMDD) 920131

3. DOCUMENT TITLE
Water Purification Unit Reverse Osmosis, 3,000 GPH

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (if applicable) (2) AUTOVON	7. DATE SUBMITTED

8. PREPARING ACTIVITY

a. NAME Carolyn B. Johnson	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (703) 704-3468
c. ADDRESS (Include Zip Code) US Army Belvoir RDE Center ATTN: STRBE-TSE Fort Belvoir, VA 22060-5606	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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